

GAIRA

Pure Ag-SERS Evaluation (V6)

A detection gate before recovery — separating measurement failure (invisible on silver) from representation failure (measured but chemistry not recovered).

Frozen atlas fingerprint 09ed804a40836f4a05a91ba10900cded · reuses V5 recovery flags unchanged

22/51 analytes pass the Stage-0 detection gate · 29 undetectable · thresholds validated before freezing
(validate_detection.ipynb)

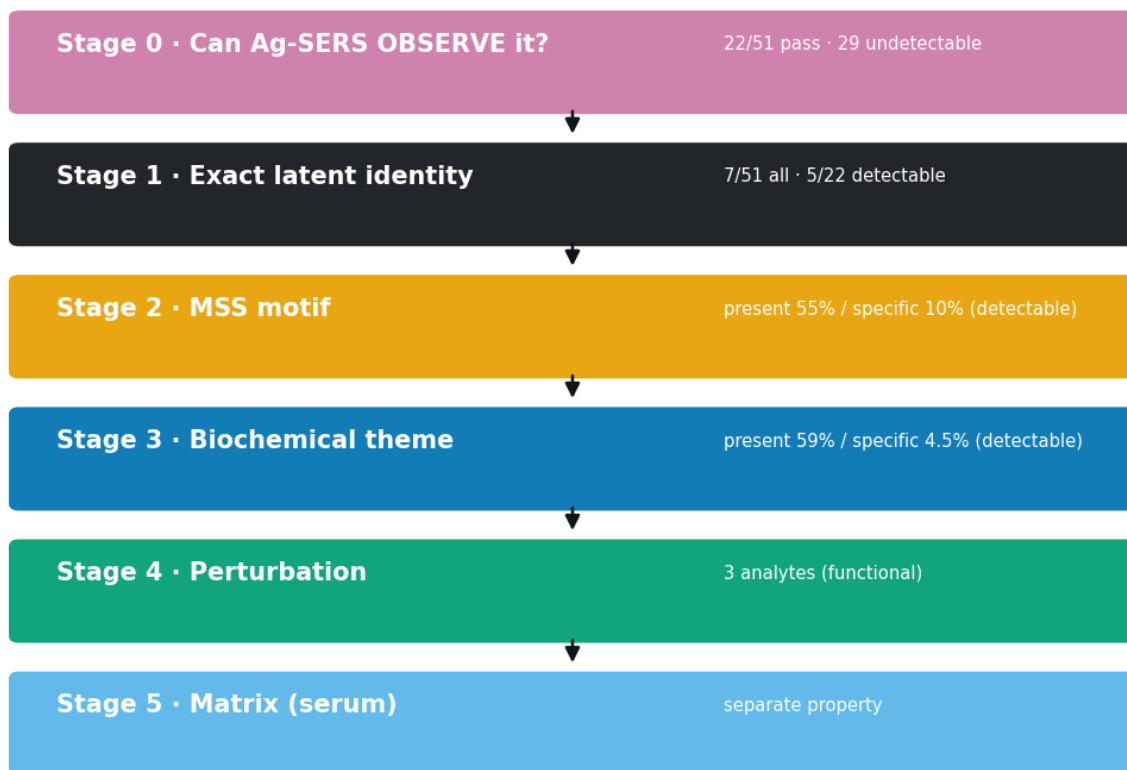
Among detectable analytes, exact identity ~doubles (14%→23%) and presence rises, but analyte-specific recovery stays low — the residual failure is representational. A learned Raman→SERS transfer model is justified for ~11 analytes; the rest need a better substrate.

1 · Executive summary & detection philosophy

V5 evaluated all 51 analytes equally — including analytes whose Ag-SERS is essentially blank. That conflates measurement failure (invisible on silver) with representation failure (measured but chemistry not recovered). V6 inserts a Stage-0 detection gate that asks a MEASUREMENT question: does this Ag-SERS spectrum contain reproducible analyte information above noise/background?

22/51 analytes pass; 29 are undetectable. Only detectable analytes are eligible for identity/motif/theme evaluation — mirroring how spectroscopy is actually done: confirm signal before interpreting it.

V6 hierarchy — Stage 0 detection gates every later stage



only DETECTABLE analytes are eligible for identity / motif / theme evaluation

Figure 1. Stage 0 gates every later stage.

2 · Detection metrics, thresholds & distribution

Deterministic weighted Detection Confidence (0–1), NO ML: replicate Pearson (0.45), Spearman (0.10), peak SNR (0.20), variance concentration (0.15), reproducible peaks (0.10). Tiers GOOD ≥ 0.65 / MODERATE ≥ 0.50 / POOR ≥ 0.40 / UNDETECTABLE < 0.40 ; pass ≥ 0.50 . Replicate cosine was rejected — baseline-inflated (0.93–0.99 for all). Validated before freezing: anchors pass (xanthine 0.99, ergothioneine 0.97, urate 0.89, adenine 0.63) and fail (glucose 0.43, tyrosine 0.42, oleate 0.33) for adsorption reasons.

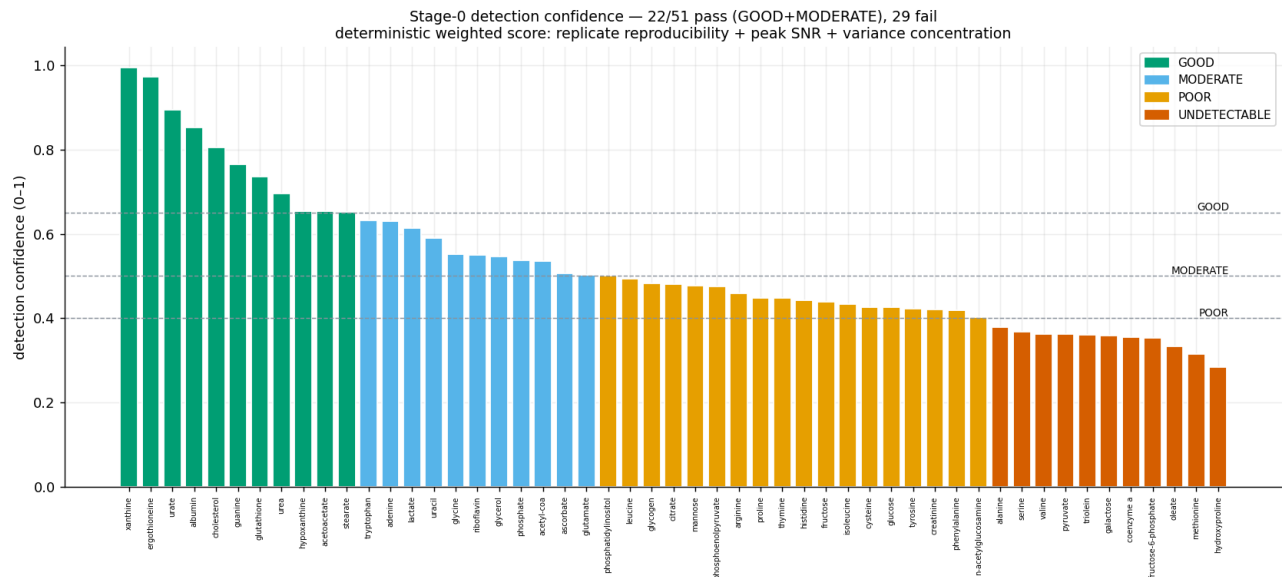


Figure 2. Detection confidence per analyte with tier bands.

3 · Representative spectra — why analytes pass or fail

Representative Ag-SERS spectra: Ag-SERS (blue) · blank (grey) · difference (red) · peaks (▼)
top row PASS (sharp reproducible peaks); bottom rows FAIL (blank-like, structureless)



Figure 3. Ag-SERS (blue) · blank (grey) · difference (red) · peaks (▼). Top row PASS (sharp reproducible peaks); bottom rows FAIL (blank-like).

4 · Hierarchical recovery — detectable only

Level	all n	all frac	det n	det frac	gain
exact analyte	7	0.137	5	0.227	0.090
NMF component	2	0.039	2	0.091	0.052
MSS present (top-3)	19	0.396	11	0.550	0.154
MSS specific	2	0.042	2	0.100	0.058
theme present (top-3)	25	0.490	13	0.591	0.101
theme specific	1	0.020	1	0.045	0.025
perturbation	3	0.059	3	0.136	0.077

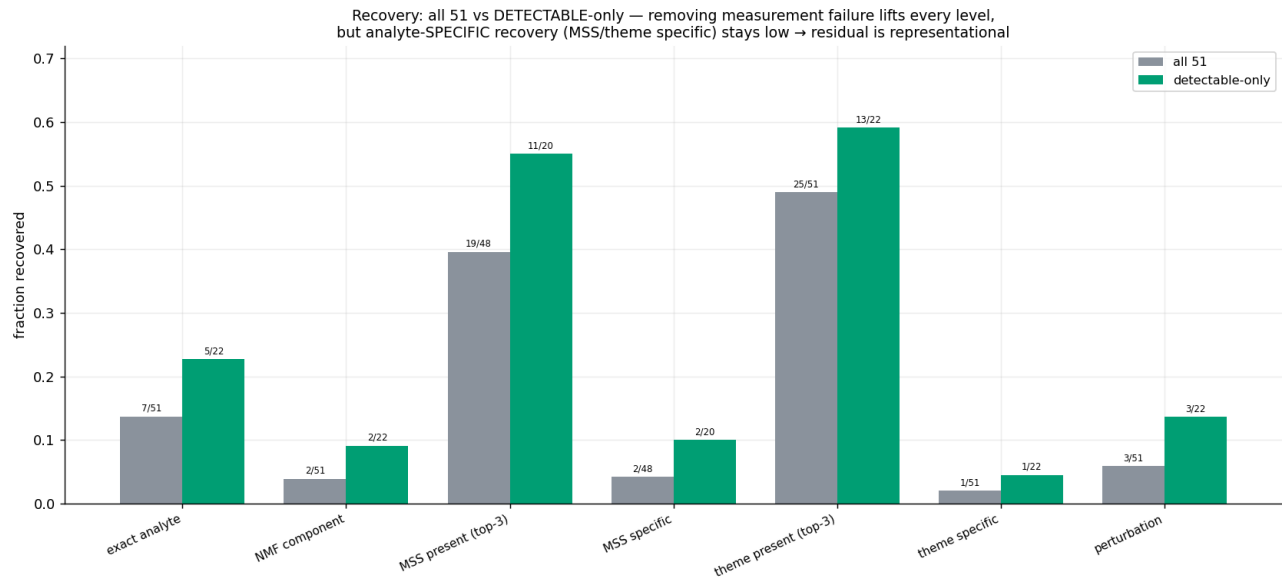


Figure 5. Removing measurement failure lifts every level, but analyte-specific recovery stays low → the residual is representational.

5 · Transfer-model decision & roadmap

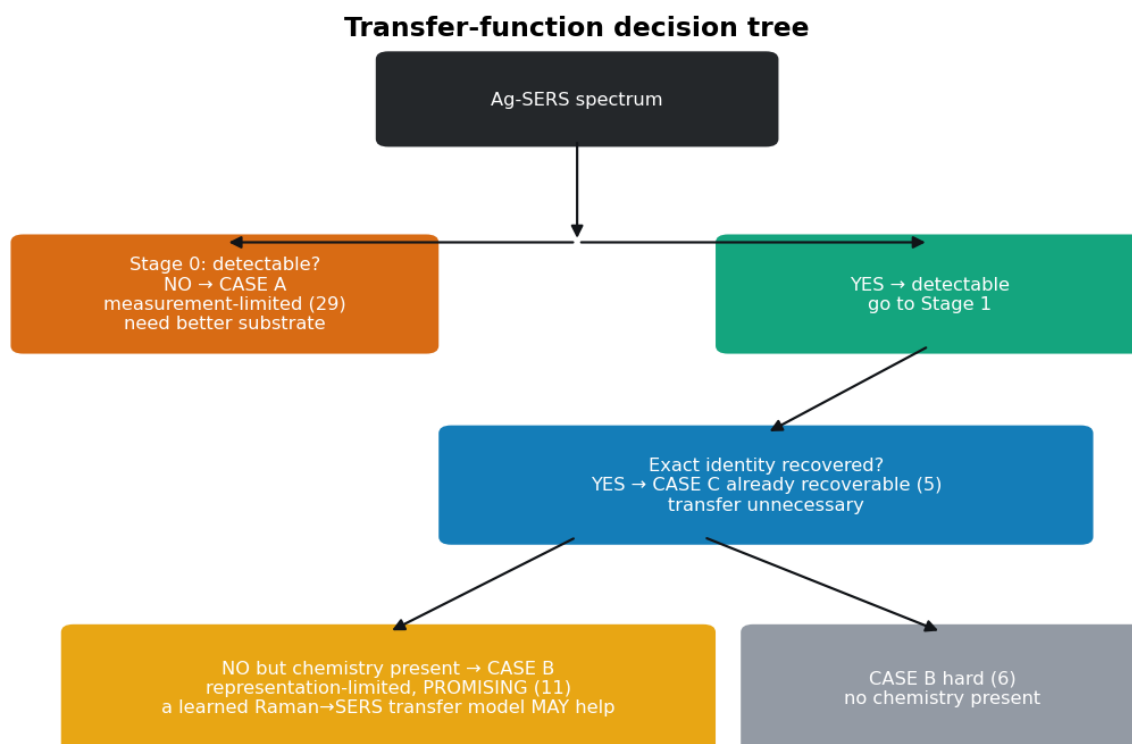


Figure 7. Decision tree: A measurement-limited · B representation-limited · C already recoverable.

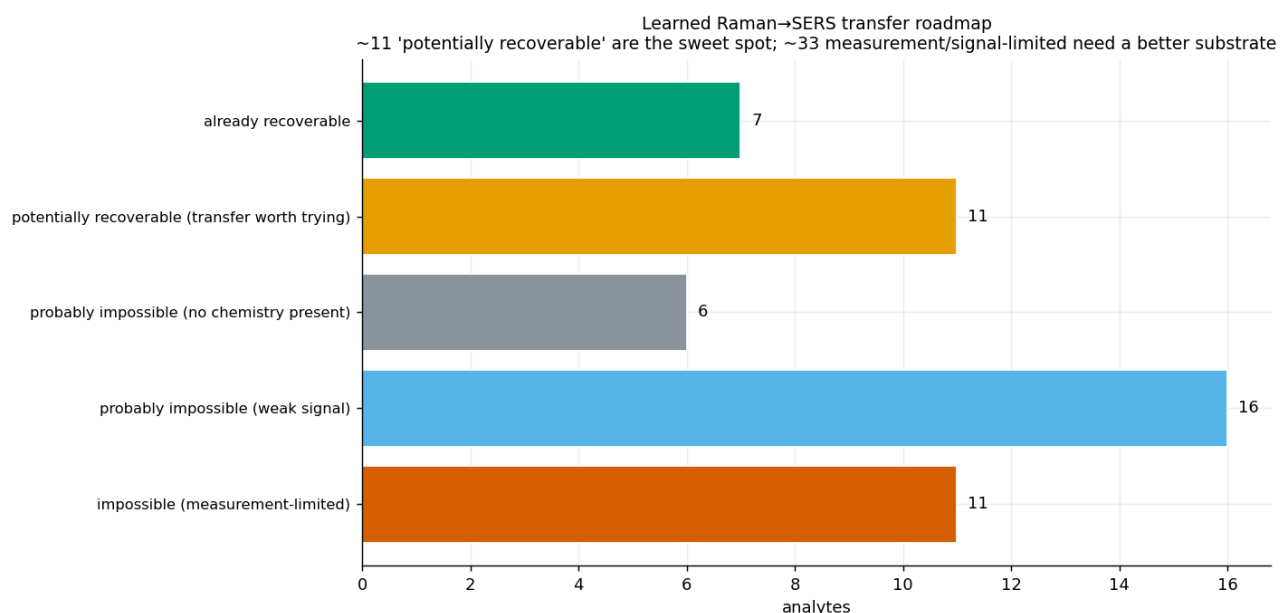


Figure 8. Learned Raman→SERS transfer roadmap — ~11 'potentially recoverable' are the target set.

6 · Recommendations, limitations & conclusions

Recommendations. Build a learned Raman→SERS transfer model on the ~11 detectable, representation-limited analytes (Case B promising); the 29 undetectable analytes need a better substrate, not a model. Extend dynamic perturbation (DART) — the only route that recovered class chemistry in V5.

Limitations. No pure Ag-colloid buffer blank (serum blank used as the Ag background); 5 replicates limit resolution; the weighting is a transparent, validated choice, not unique; the gate is conservative (creatinine and thymine were identity-recovered yet fall just below it); Raman-trained atlas; no learned modality model yet.

Conclusions. A measurement gate before interpretation cleanly separates 'we can't see it' from 'we can't recover it.' Half the analytes are simply invisible on this substrate; among the visible ones recovery improves but specific chemistry still largely fails to transfer — a real representation gap that a learned model could target for ~11 analytes. Frozen atlas 09ed804a40836f4a05a91ba10900cded unchanged; reuses V5 recovery flags; deterministic.